

1. There should be 3 rectangles with width $\frac{6-0}{3}$ each touching the curve f with its *upper right corner* at the values $f(2)$, $f(4)$, $f(6)$.
2. $2 \cdot (20 + 16 + 12)$
3. $\frac{1}{2}3 \cdot 6 - \frac{1}{2}1 \cdot 2 = 8$
4. $5 + (-7) = -2$
5. $5 - (+2) = 3$
6. a. $x^9 + C$
b. 1
7. $\frac{x^3}{3} - 4x^2 + x$
8. a. $\frac{1}{3}\sin 3x + C$
b. $0 - 0 = 0$
9. $-\frac{1}{x} + \frac{2}{3x^{3/2}} + C$ (break up the fraction)
10. use $u = 3x^3 + 1$
a. $\frac{1}{45}(3x^3 + 1)^5 + C$
b. $\frac{3254867}{15}$
11. $\frac{1}{3}\sin^3 \theta + C$ (substitution)
12. $-\sqrt{4-t^2} + C$
13. 3250 feet
14. $\frac{2}{3}(t+9)^{3/2} + 2$
15. a. $s(t) = 20t - 5t^2$
b. 40 (distance up+distance down)
16. $\frac{125}{6}$
17. 16
18. $1 + 1 = 2$
19. $\frac{81}{2}\pi$
20. $\frac{\pi}{2}$
21. $\frac{335}{108}$ (substitution)
22. 12 (solution on Jenzabar/handouts)
23. $\frac{1}{x}$
24. $3e^x \ln x + \frac{3e^x}{x}$ (product rule)
25. $-\frac{\sin x}{\cos x} - 2$ (chain rule)
26. $(-2 + 2x)e^{-2x+x^2}$ (chain rule)
27. $\frac{e^t}{(e^t + 1)^2}$ (quotient rule)
28. $-e^{-3x} + \ln|x| + C$
29. $(-\frac{x}{4} - \frac{1}{16})e^{-4x} + C$
30. $\frac{\cos 4x}{16} + \frac{x \sin 4x}{4} + C$
31. $-\frac{4 \ln x + 1}{4x^4} + C$
32. $3(x-1)e^x|_0^{\ln 1} = 0$
33. $x \ln 3x - x + C$
34. $\frac{x}{2} - \frac{\sin 6x}{12} + C$
35. $\sin x - \frac{\sin^3 x}{3} + C$
36. $-\frac{1}{4}x - \frac{1}{8}\sin 4x + C$
37. $\frac{\sin^2 x}{2} - \frac{\sin^4 x}{4} + C$
38. $\sin^{-1}\left(\frac{x}{2}\right) + C$
39. $4 \sin \theta|_0^{\pi/2} = 4$
40. $8 \sin^{-1}\left(\frac{x}{4}\right) + x\sqrt{16-x^2} + C$